



Commercial Integrated Circuits





Available ICs in digital systems

Logic functionalities are implemented in ICs as:

- Standard logic (gates, flip-flops, etc)
- Programmable logic (Field Programmable Gate Arrays – FPGAs)
- Application Specific Integrated Circuits (ASICs)

Standard logic has been developed through many **Logic families** (either CMOS and TTL), each family being characterized by a different IC configuration and related different speed performance and power consumption



ICs in digital systems

In the past a system was composed by several different ICs, one for each specific function:

- Logic gates
- FF
- Buffers
- Adders
- Counters
- ...



ICs in digital systems

Today, most classic standard logic circuits are obsolete: they can be replaced by a single programmable PLD (FPGA) containing thousands of gates:

- Logic gates, FF
- Multipliers, Adders
- Comparators, encoders, multiplexers
- Clock managing circuits (PLL)
- ...



ICs in digital systems

Integrating digital logic in a single device yields:

- Lower PCB dimensions
- Lower parasitics
- Higher velocities (eg: 500 MHz or more)

Only buffers, registers, and single gates,
“survive” as Glue logic



ICs in digital systems

Typical systems configuration include:

- Programmable logic (FPGA, uC, DSP)
- Glue logic
- Front-end ICs (ADC / DAC, I/O functions)
- Clock system





Packaging of ICs

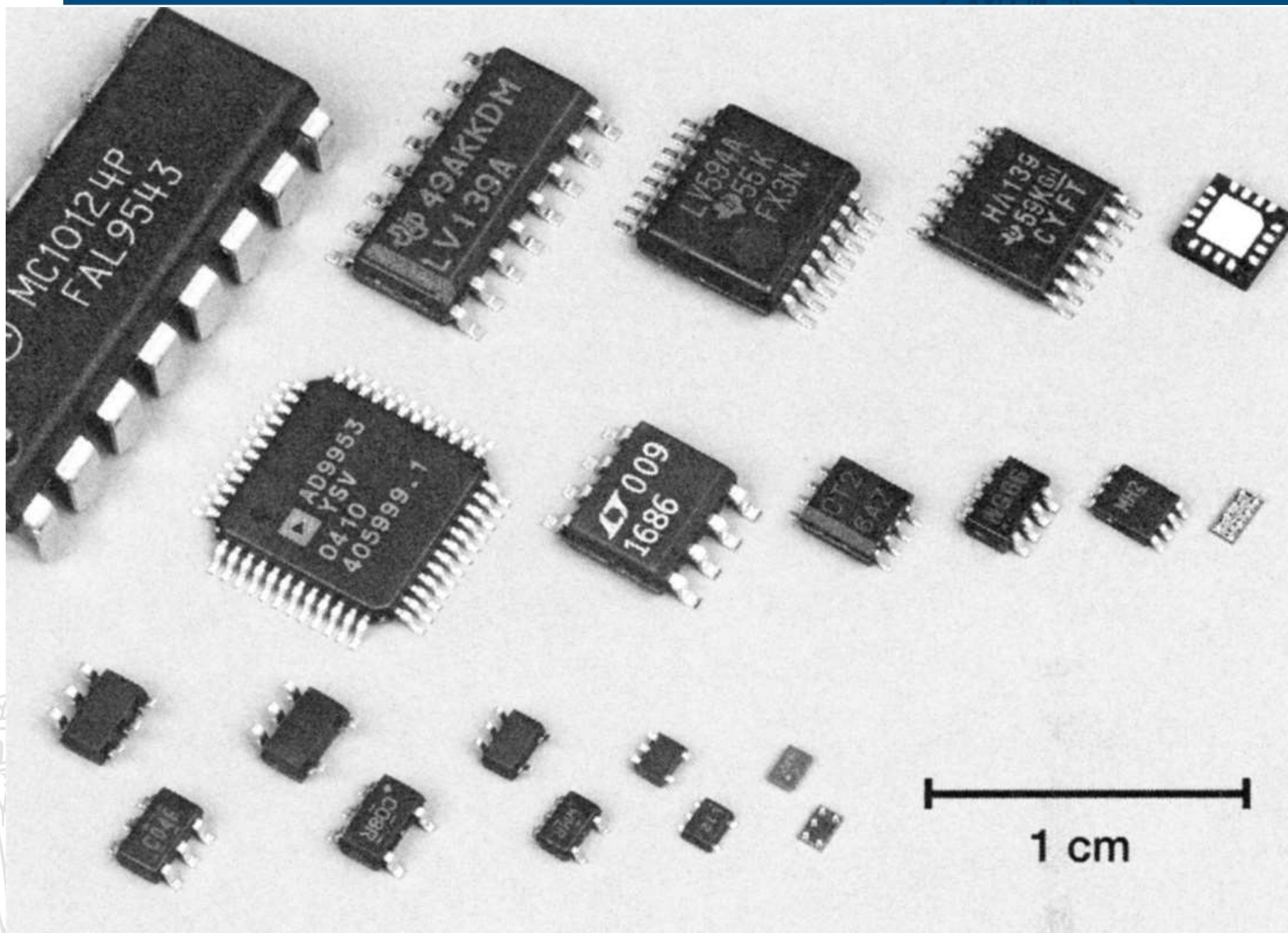
The package of ICs is important because:

- it determines the amount of occupied area
- It unavoidably involves parasitic effects (input capacitance and inductance)

In general, the smaller the package, the better it is for both previous points.

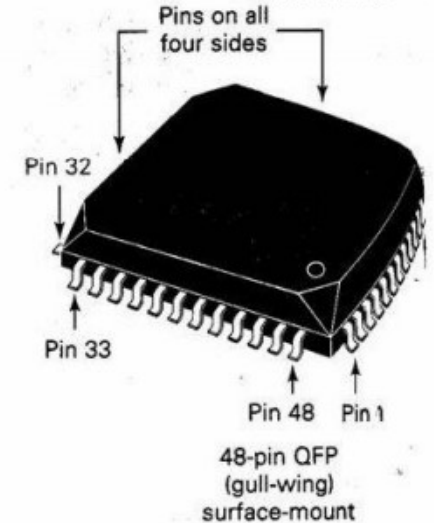
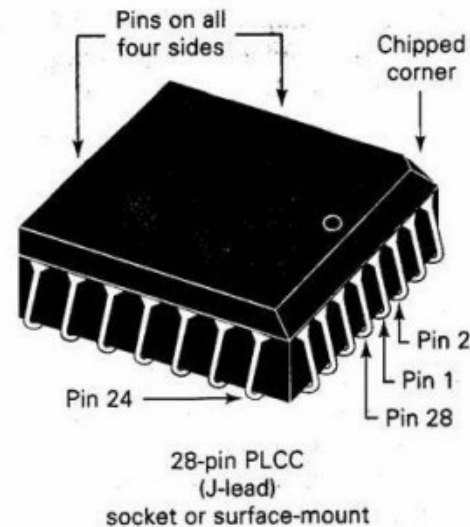
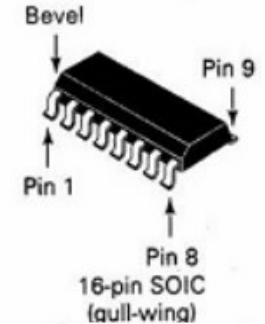
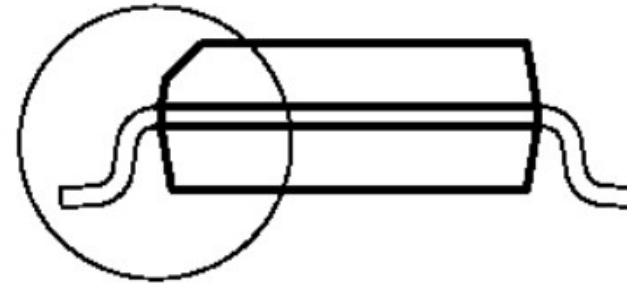
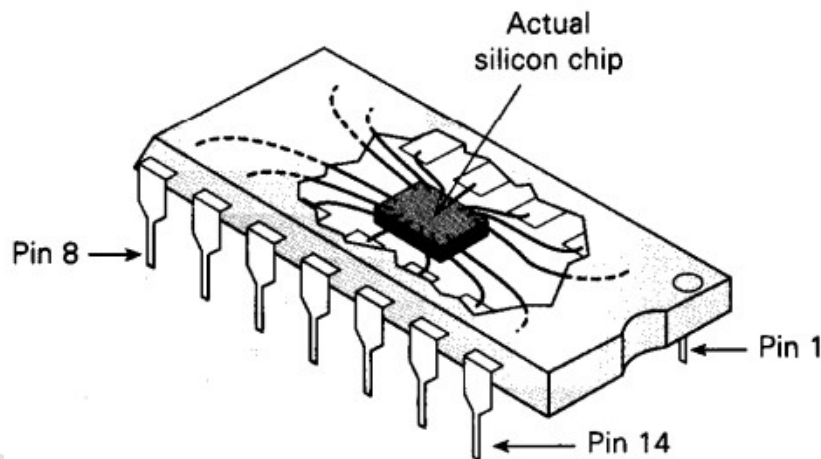
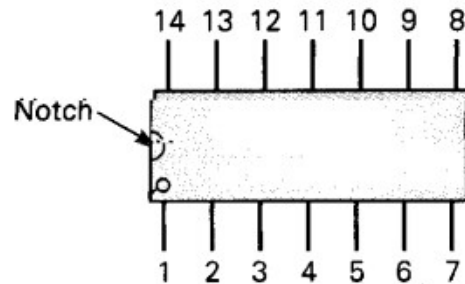
There are several types of package:

- Dual-in-line package (DIP: used in first digital ICs)
- Surface mount device (SMD) package (currently mostly used for general purpose devices)
- Ball grid arrays (BGA) package (for large devices with many I/O pins)
- Bare die (for ASICs: devices directly connected to the PCB; eg: wire bonding)



Horowitz-Hill: Arte dell'elettronica, Zanichelli (2018)

ICs: DIP – smd packages

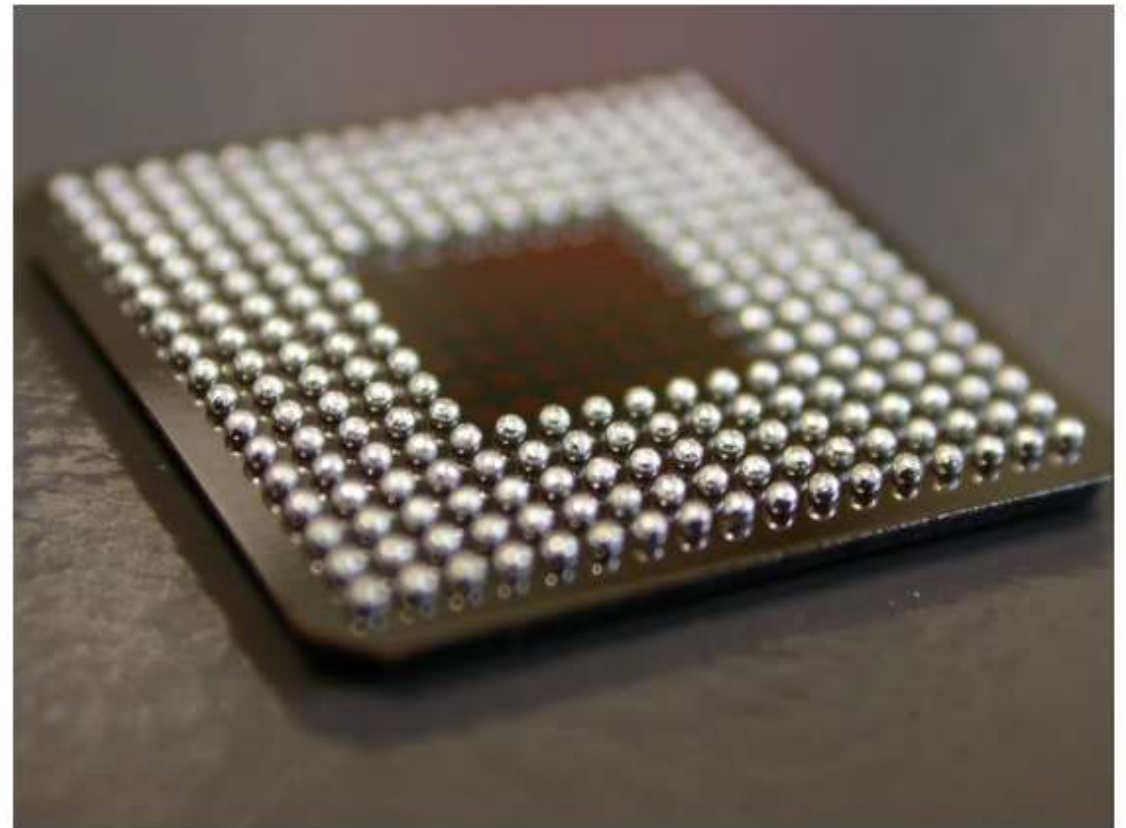
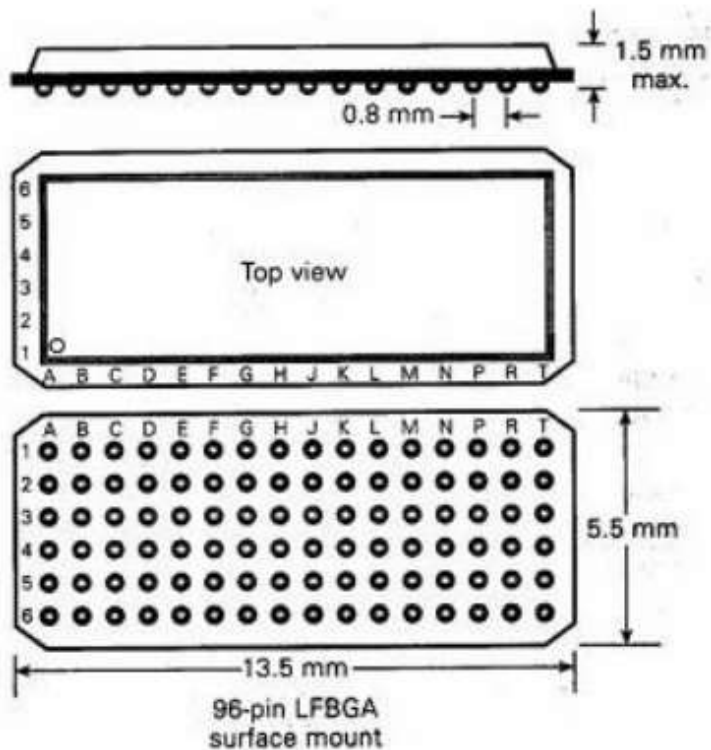


DIP (Dual-In-Line)

Surface Mount Devices (smd)

Limitation: n. of pins that can stay on the borders is low

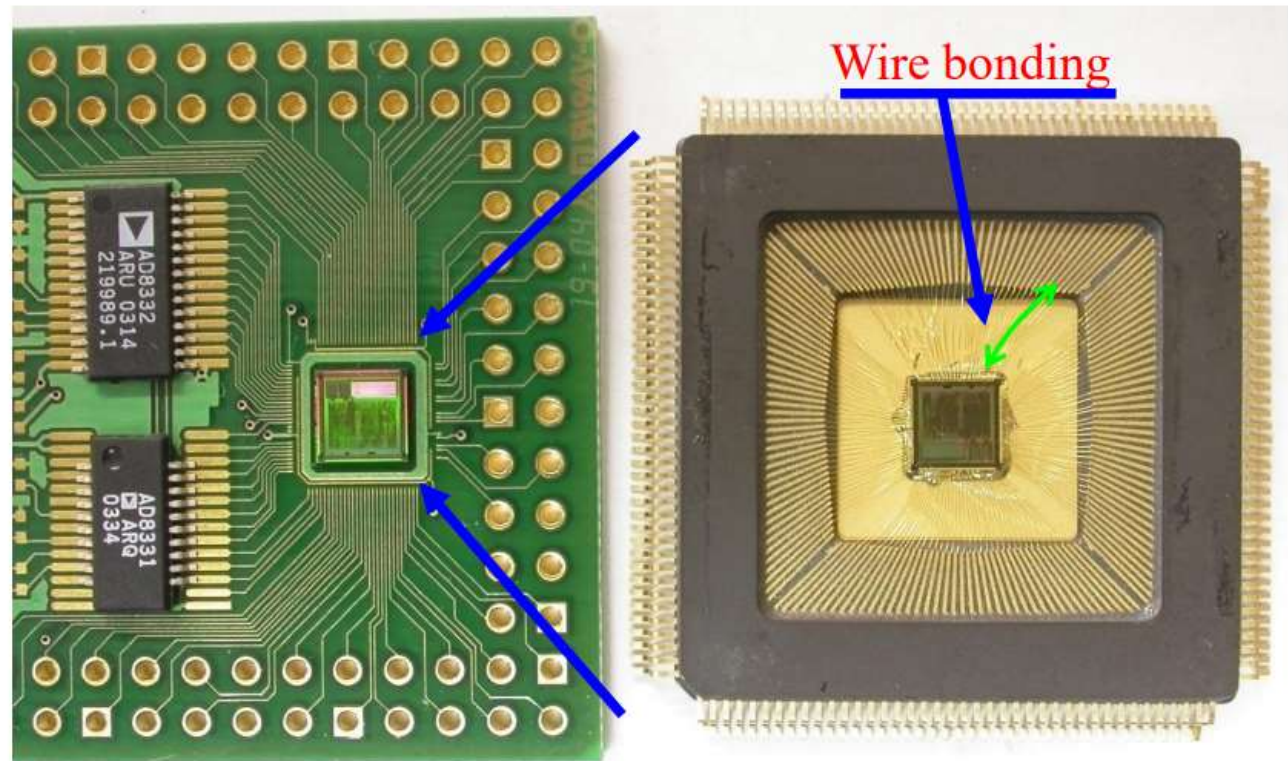
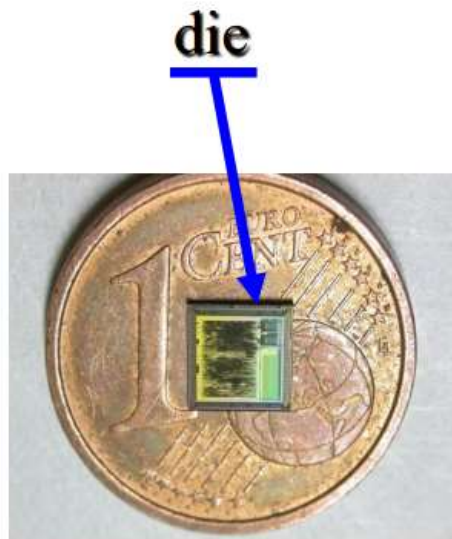
Ball Grid Array (BGA) package



Pins distributed on a surface → lower dimensions → lower parasitics → higher speed

Advanced packaging techniques

- Wire bonding or flip-chip can be applied to bare die over PCB





SMT MANUFACTURING IN
JLCPCB FACTORY

UNA VOLTA SOTTO I COMPONENTI
N. PASTA D'ATTACCO

30/04/2026

=> FA GOMIT

=> SOLO PER IL MONTAGGIO

Standard logic families

TTL – 5V

CMOS – 5V

Low Voltage CMOS (<5V)





5V logic families

- **TTL/L*S** standard/Low power Schottky → ~~standard~~ ~~Schottky~~ *obsolete*
- **ALS** Advanced LS → *Divorzio la Petrucci* compared to LS: 0.5×Power, 2 × Speed
- **F** Fast → max speed, high power consumption

Note: In Low power series, power consumption is limited by using higher resistance values, at the expense of speed

The TTL family is not used today.

A few F devices are used only when equivalent CMOS are not available.

- **AHC** Advanced HighSpeed CMOS *general purpose logic*
- **AC** Advanced CMOS *max speed*

non è HIGH SPEED → ADVANCED CMOS è più veloce



CMOS Logic Families

AC (Advanced CMOS)

- Vcc: 5 V
- High speed
- Pure CMOS levels
- Legacy, not TTL compatible

AHC (Advanced High-Speed CMOS)

- Vcc: 2–5.5 V
- High speed
- CMOS levels
- More flexible than AC

LV (Low Voltage)

- Vcc: 3.3 V
- Moderate speed
- Early low-voltage family

LVC (Low Voltage CMOS)

- Vcc: 1.65–3.6 V
- High speed
- 5 V tolerant inputs
- Widely used

ALVC (Advanced Low Voltage CMOS)

- Vcc: 1.65–3.6 V
- Very high speed
- Optimized for performance

AVC (Advanced Very Low Voltage CMOS)

- Vcc: 1.2–3.6 V
- Very high speed
- For modern low-voltage cores

AUC (Advanced Ultra Low Voltage CMOS)

- Vcc: 0.8–2.7 V
- Extremely high speed
- Ultra-low power

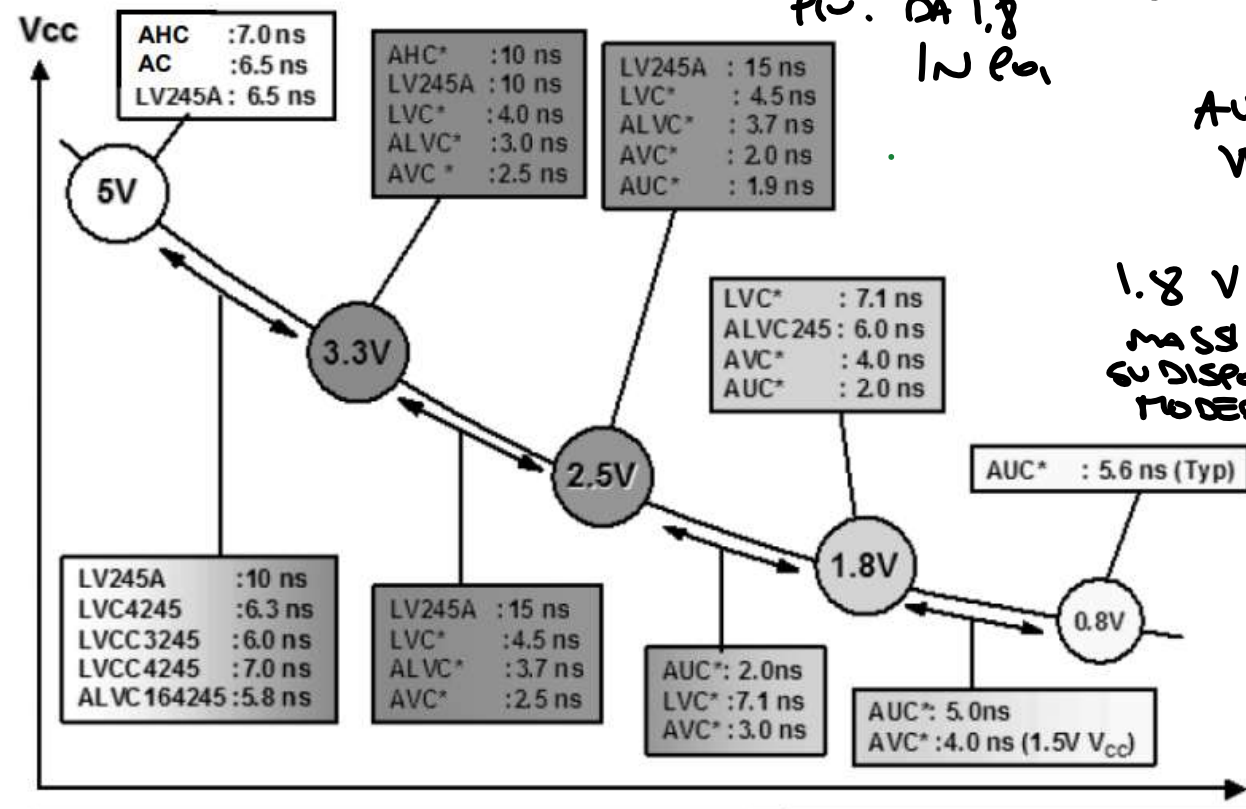
↳
QUESTE
ATTUALI

CMOS Logic Families- Comparison

- 5V families are now obsolete *non si +20V A*
- New designs are based on low power CMOS logic circuits (3.3V, but also: 2.5V, 1.8V, 1.2V...) *SE DIMINUIAMO LA TENSIONE DELL'ALIMENTAZIONE CIRCUITO SI CALDA PIÙ PICCOLO → AUMENTO DI VELOCITÀ*

By reducing the **nominal** V_{cc} , the delay times are reduced ($\rightarrow$ higher speed)

By reducing the **actual** V_{cc} , the delay times are increased ($\rightarrow$ lower speed)



ANCHE 2.5 non c'è più da 1.8 in poi

RIDURRE POTENZA $\neq$ AUMENTO VELOCITÀ

1.8 V. massima Tensione su dispositivi moderni.

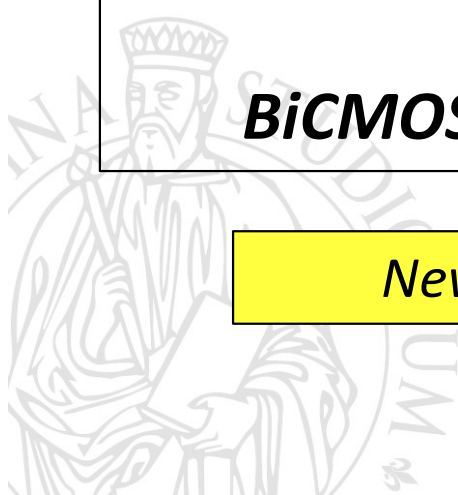


CMOS Logic Families -Summary

Family	V _{cc} Range	Speed	Key Feature
AC	5 V	High	Legacy CMOS
AHC	2–5.5 V	High	Flexible supply
LV	3.3 V	Medium	First low-voltage
LVC	1.65–3.6 V	High	5V tolerant inputs
ALVC	1.65–3.6 V	Very High	Performance optimized
AVC	1.2–3.6 V	Very High	Low-voltage systems
AUC	0.8–2.7 V	Ultra High	Ultra-low voltage

- ## GENERAL PURPOSE

- BiCMOS*** are mainly used for high-speed bus management



Bicmos

New



Acronyms interpretation

SU ~~PIÙ~~ ~~CHIAVI~~ ~~ACQUA~~
~~DIGITALE~~ ~~NUOVE~~

Each acronym should specify:

- Logic family and series (TTL, CMOS, BiCMOS; 5Volt, LV, etc)
- Functionality (AND, OR, FF, Counter, PLD...)
- Package
- Temperature range

Example: standard Texas Instruments nomenclature:

SN74AC00-xxx

SN: manufacturer (TI)

74: commercial circuits (military circuits: 54)

AC: Advanced CMOS family (other acronyms for other families)

00: identifies the logic functionality (eg: 00 → 4 NAND gates)

xxx: identifies the package (ex: D → small outline IC)



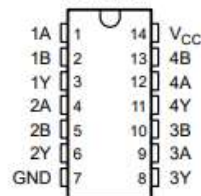
SN74ALVC00
QUADRUPLE 2-INPUT POSITIVE-NAND GATE
SCES115G-JULY 1997-REVISED AUGUST 2004

FEATURES

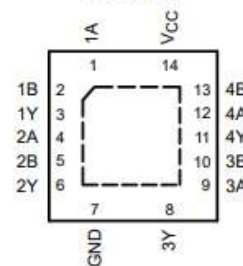
- Operates From 1.65 V to 3.6 V
- Max t_{pd} of 3 ns at 3.3 V
- ± 24 -mA Output Drive at 3.3 V $\rightarrow 33$
- Latch-Up Performance Exceeds 250 mA Per JESD 17

- ESD Protection Exceeds JESD 22
 - 2000-V Human-Body Model (A114-A)
 - 200-V Machine Model (A115-A)
 - 1000-V Charged-Device Model (C101)

D, DGV, NS, OR PW PACKAGE
(TOP VIEW)



RGY PACKAGE
(TOP VIEW)



DESCRIPTION/ORDERING INFORMATION

This quadruple 2-input positive-NAND gate is designed for 1.65-V to 3.6-V V_{CC} operation.
The SN74ALVC00 performs the Boolean function $Y = \overline{A} \cdot \overline{B}$ or $Y = \overline{A + B}$ in positive logic.

ORDERING INFORMATION

T_A	PACKAGE ⁽¹⁾		ORDERABLE PART NUMBER	TOP-SIDE MARKING
-40°C to 85°C	QFN - RGY	Tape and reel	SN74ALVC00RGYR	VA00
	SOIC - D	Tube	SN74ALVC00D	ALVC00
		Tape and reel	SN74ALVC00DR	
	SOP - NS	Tape and reel	SN74ALVC00NSR	ALVC00
	TSSOP - PW	Tape and reel	SN74ALVC00PWR	VA00
	TVSOP - DGV	Tape and reel	SN74ALVC00DGVR	VA00

(1) Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

FUNCTION TABLE
(each gate)

INPUTS		OUTPUT Y
A	B	
H	H	L
L	X	H
X	L	H

LOGIC DIAGRAM, EACH GATE (POSITIVE LOGIC)



Device description

General Features

Pin-out

Available versions

Truth-table and
Logic diagram



Comparison between logic families

↳ IN TERMINI DI PROPAGAZIONE

Comparison criteria (performance)





Performance of logic ICs

The performance of a logic IC is mainly measured in terms of:

- **Velocity** (I/O delay, max clock frequency)
- **Drive capability** (capability of providing current adequate to the needs of load ICs – key problem in driving low impedances like a bus) *⇒ BUFFER IN OUTPUT*
- **Power consumption** (always – but more markedly in CMOS circuits – depending on the frequency) *• CAPACITÀ PARASSITA*

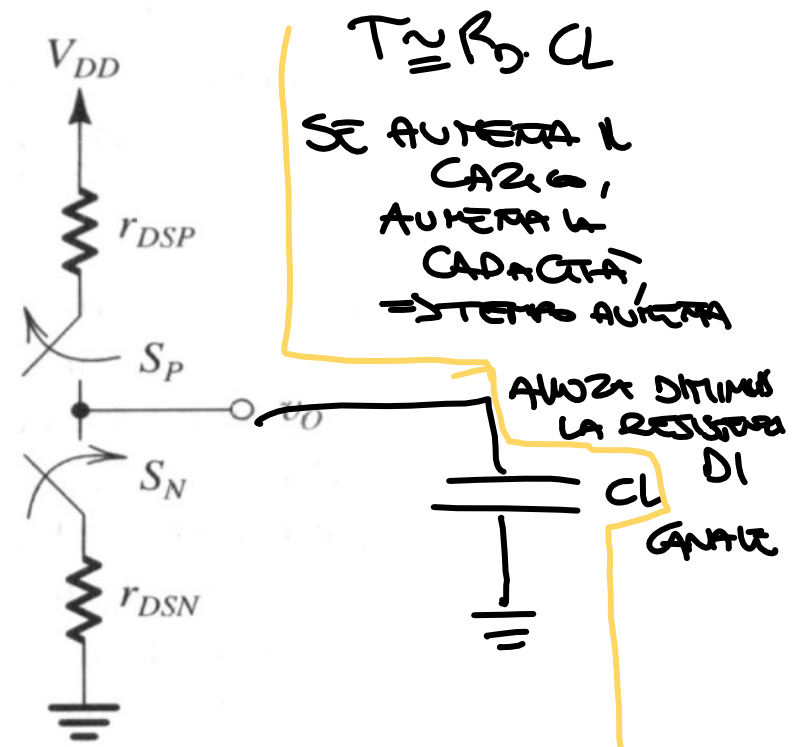
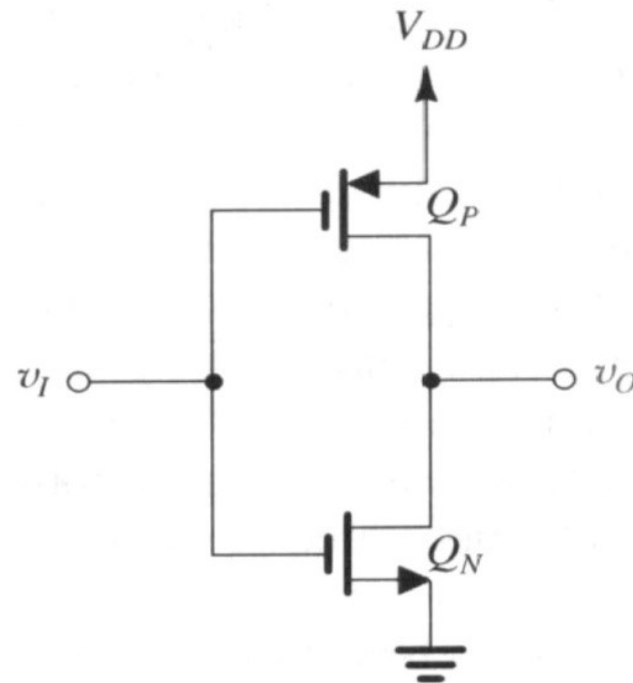
↳ DIFENDE DA → *• FREQUENZA DEL VANTO*
• AUMENTAZIONE DELLA TENSIONE

In general, each family offers a different **compromise** among such characteristics

Usually more speed obtained at expense of more power

Speed vs power

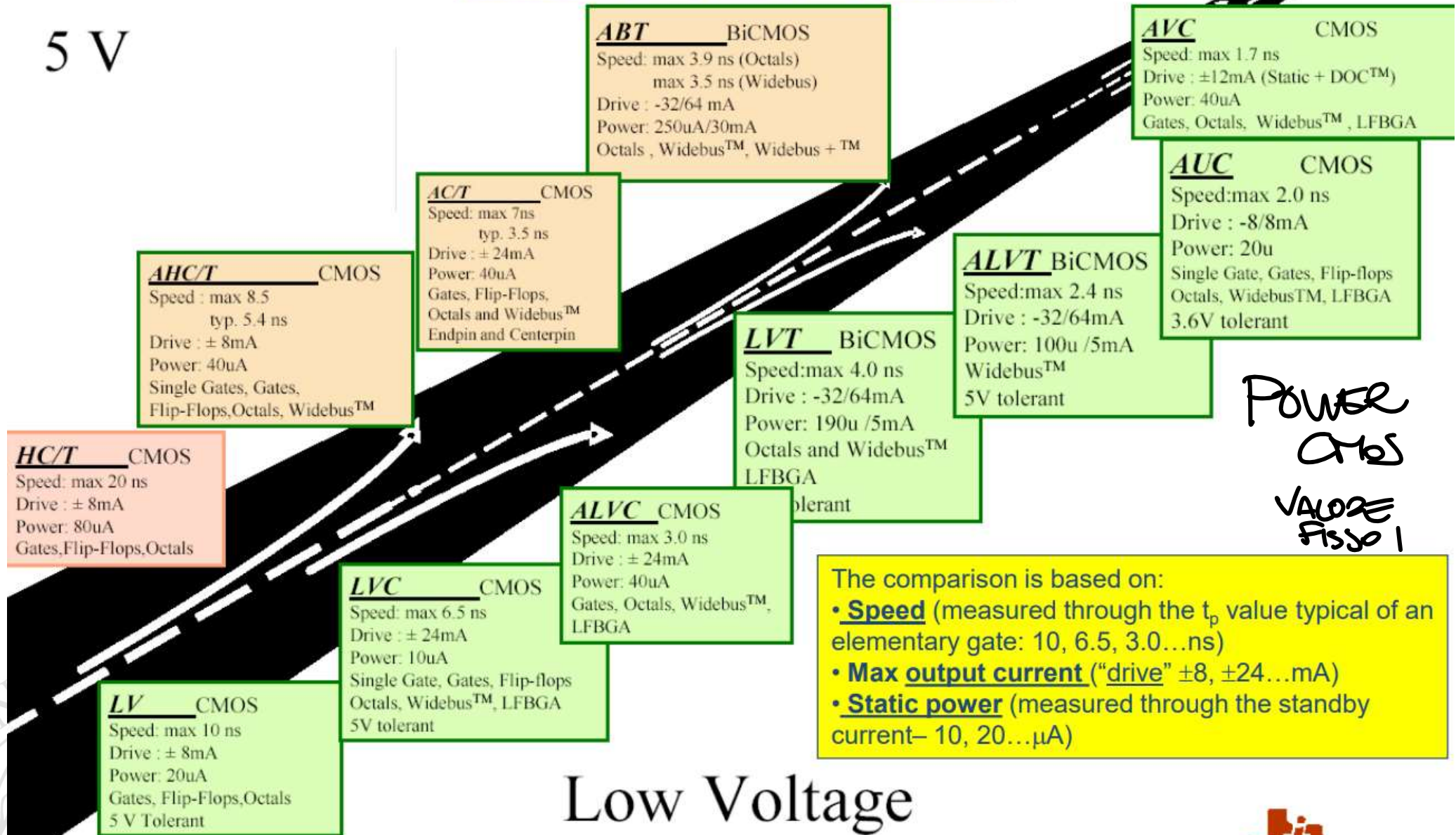
NOT CMOS



CROSS-CONDUZIONE $\rightarrow$ C. TA IN SUI E VDS. PASSA TANTA CORRENTE

Logic Family Comparison

5 V





Digital integrated circuits

Interconnection issues: I/O voltages
Output characteristics
Datasheet content

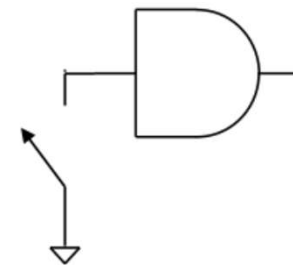


ICs: How to set an input value

In ICs, the input voltage, equivalent to a logic 1 or 0, can be set through a switch.

How?

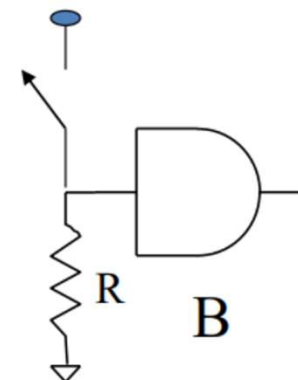
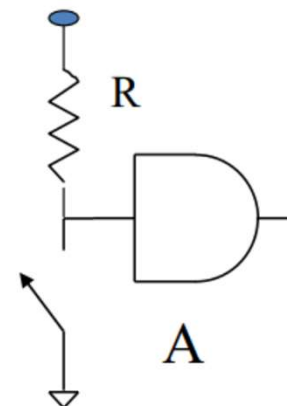
Let's first assume that 1 corresponds to V_{cc} and 0 to Gnd (0V). The switch must NOT be connected in this way → because it works well for a low input level, but the input remains floating (ie, undefined) when the switch is off. (at least in CMOS-input families)



The following approaches are more correct:

- A:** switch ON → Input: 0V
switch OFF → Input: $\approx V_{cc}$
- B:** switch ON → Input: V_{cc}
switch OFF → Input: $\approx 0V$

This is fully true only if the IC input does not sink any current. In the following, the importance of IC input (and output) currents and voltages, and their reciprocal influence, will be highlighted.





Unused input pins:

CMOS input pins are very high impedance (the gate is insulated):

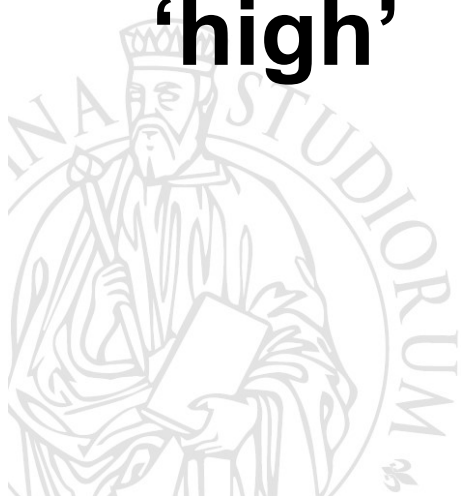
- a 'free' input pin can spontaneously charge at any voltage between 0 and V_{dd}
- At intermediate (not valid logic) voltages both NMOS and PMOS are in conduction → possible cell damage



Unused input pins:

TTL input pins are the emitter of a BJT:

- The corresponding base is pulled high internally
- Unconnected input pins are pulled high by the internal base pull-up and are **seen as logic 'high'**





I/O voltage/current levels

- Each output (driver to an input) must **guarantee** correct (low/high) voltage levels, as requested (considered “**admissible**”) by load ICs
- Each output must **guarantee** (low/high) output currents adequate to the load conditions
- When different logic series or technologies are interconnected, their I/O **compatibility** must be checked



I/O voltage levels: compatibility

When ICs are interconnected:

- The (voltage) compatibility is ensured when the *guaranteed output range* is included in the *admissible input range*. In practice, when:

$$V_{OHmin} > V_{IHmin}$$

$$V_{OLmax} < V_{ILmax}$$

- **Noise Margins:**

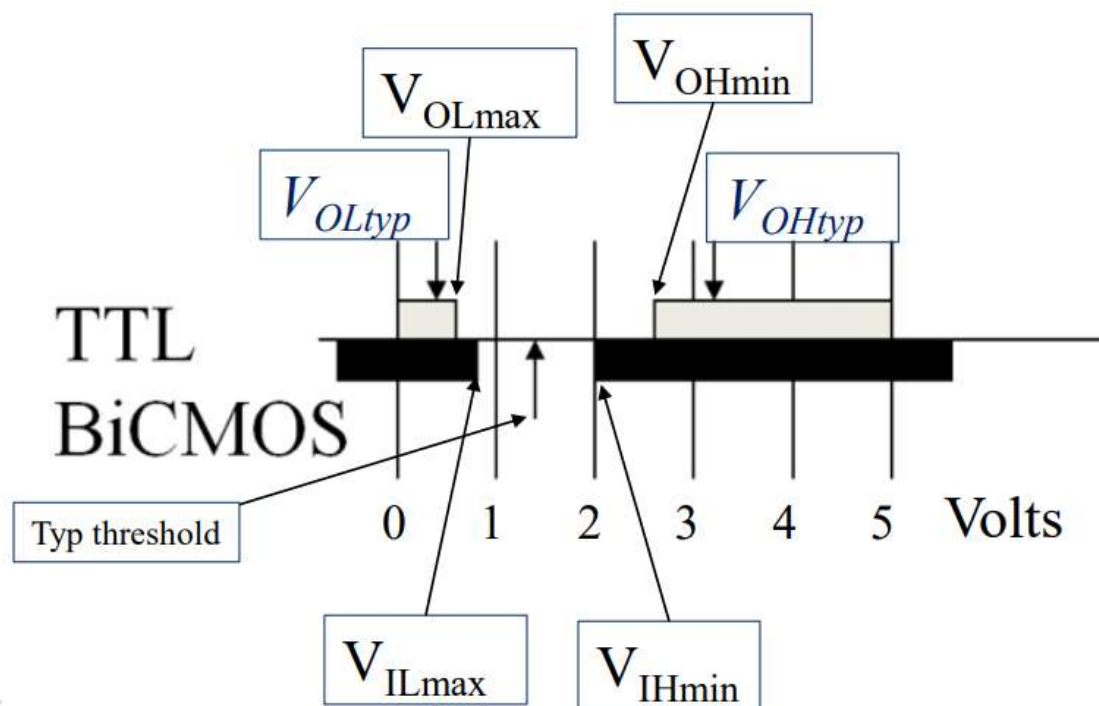
$$NMH = V_{OHmin} - V_{IHmin}$$

$$NML = V_{ILmax} - V_{OLmax}$$

Worst case design:

Recommended design methodology that makes reference to **Min/Max admissible values** (not to *typical* values)

I/O voltage levels: compatibility



Within the TTL and BiCMOS families, compatibility is ensured since the guaranteed output range is within the admissible input range

Note: The modified input of BiCMOS families makes them TTL compatible!

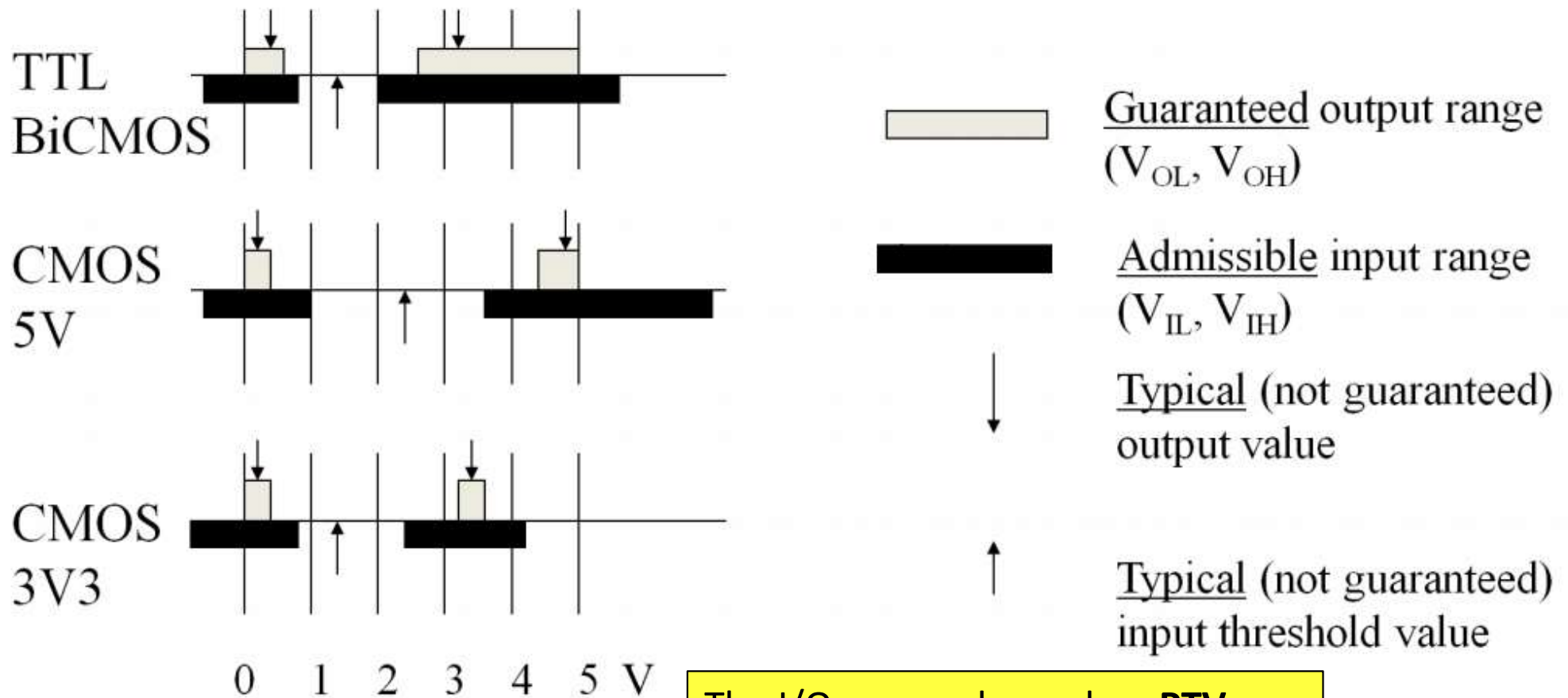
The typical threshold value is purely indicative. The manufacturer only guarantees that it is between V_{ILmax} and V_{IHmin}

Guaranteed output range (V_{OL}, V_{OH})

Admissible input range (V_{IL}, V_{IH})



I/O voltage levels

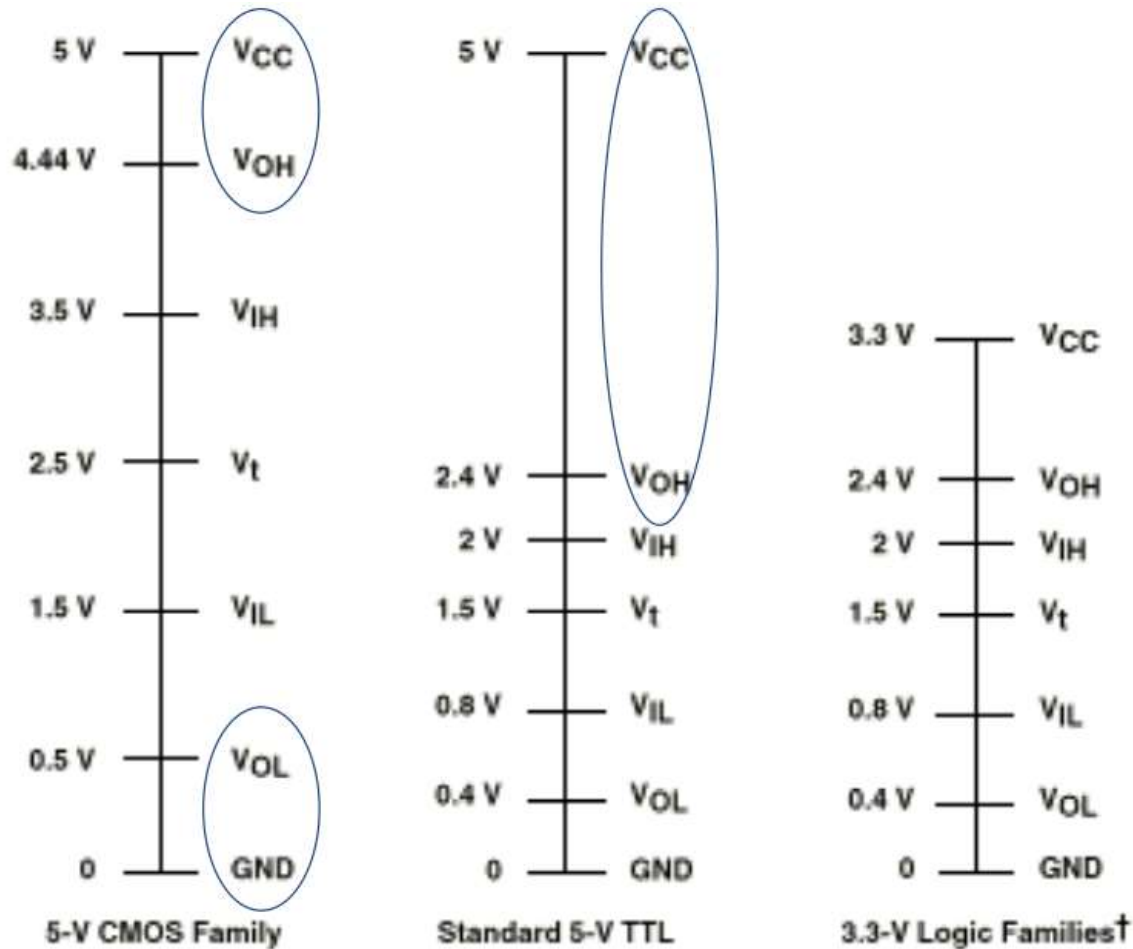


The I/O ranges depend on **PTV**:

- Manufacturing Process
- Temperature
- Voltage (V_{CC})



Compatibility: 5V CMOS – TTL – 3.3V CMOS



TTL and 3.3V CMOS are directly compatible
TTL and 5V CMOS are not!



Compatibility: 5V CMOS – TTL

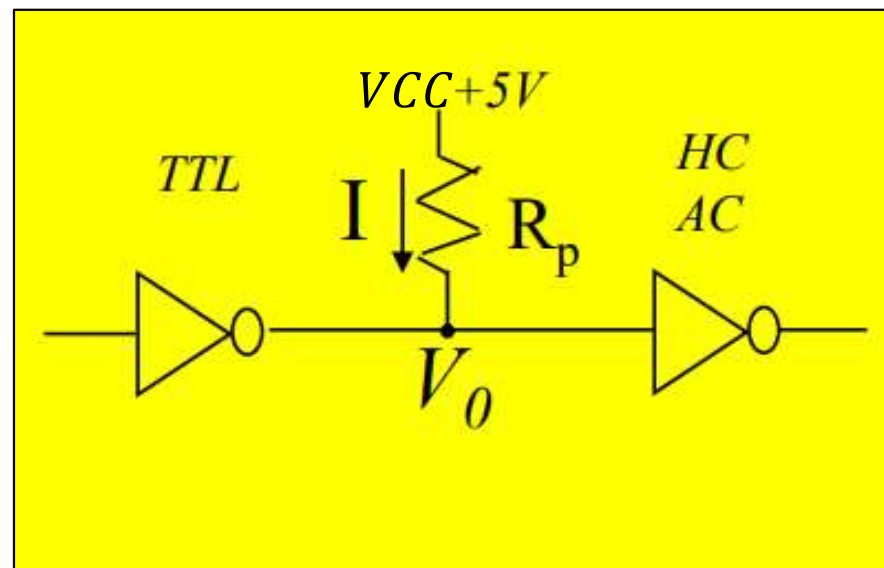
Critical case: TTL IC driving a CMOS powered at 5V:

$V_{OHmin} > V_{IHmin}$ is not guaranteed

Possible solution: **Introducing a pull-up resistor, R_p**

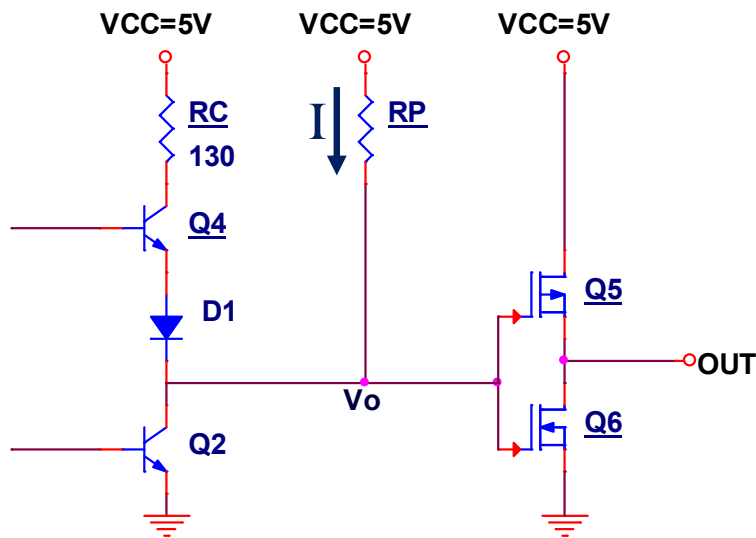
R_p will need to provide some I_{OH} to the CMOS in order to effectively pull-up the input in a reasonably short time. Moreover, with output low, the current should be maintained at **safe level** for the TTL output stage

How to set R_p ?



Compatibility: 5V CMOS – TTL

V_{OH} : The pull-up resistor, R_p , must be able to bring the output voltage of the TTL port to values greater than 3.5V



$$V_O = V_{CC} - R_P \cdot I > 3.5V \Rightarrow R_P < \frac{V_{CC} - 3.5V}{I} \Rightarrow R_P < \frac{1.5V}{I}$$

Static Condition:

$$I_{IHCMOS} \cong 0 \Rightarrow V_O = V_{CC} - R_P \cdot I \cong V_{CC} = 5V$$

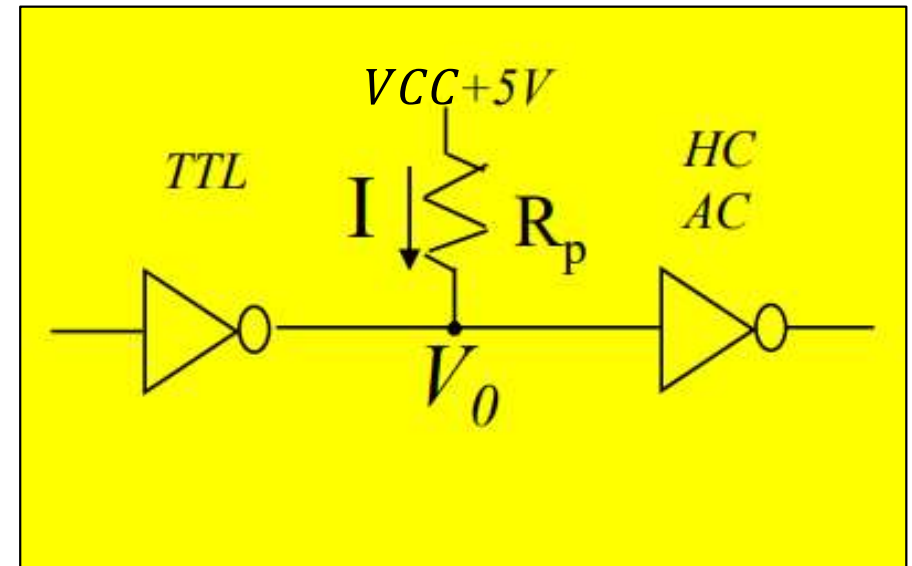
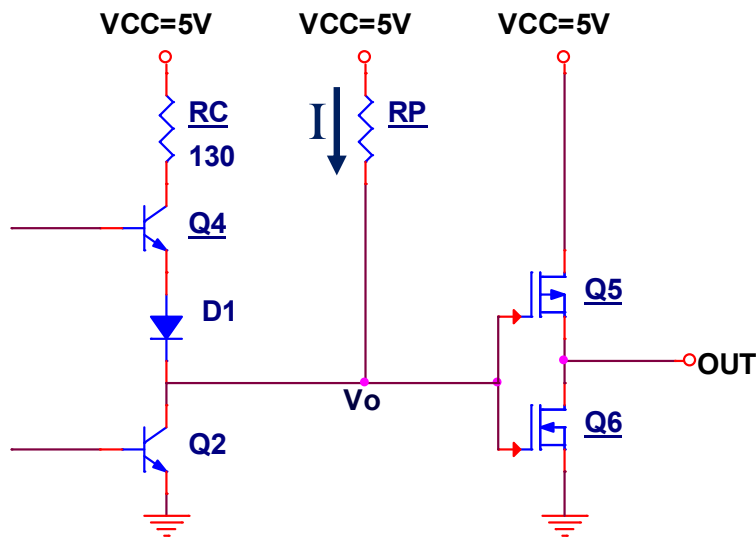
Dynamic Condition:

$$I_{OHTTL}^{MAX} = 1mA \Rightarrow R_P < \frac{1.5V}{I_{IOHTTL}^{MAX}} = \frac{1.5V}{1mA} \Rightarrow R_P < 1.5k\Omega$$

Compatibility: 5V CMOS – TTL

VOL: $V_{OLTTL}^{MAX} = 0.5V < V_{ILCMOS}^{MAX} = 1.5V$ There is compatibility.

I flows inside the TTL port.



$$I_{OLTTL}^{MAX} = 10mA \Rightarrow V_O = V_{OLTTL}^{MAX} = 0.5V \quad e \quad I < I_{OLTTL}^{MAX} = 10mA$$

$$I = \frac{VCC - V_O}{R_P} = \frac{VCC - V_{OLTTL}^{MAX}}{R_P} < I_{OLTTL}^{MAX} \Rightarrow R_P > \frac{VCC - V_{OLTTL}^{MAX}}{I_{OLTTL}^{MAX}} \Rightarrow$$

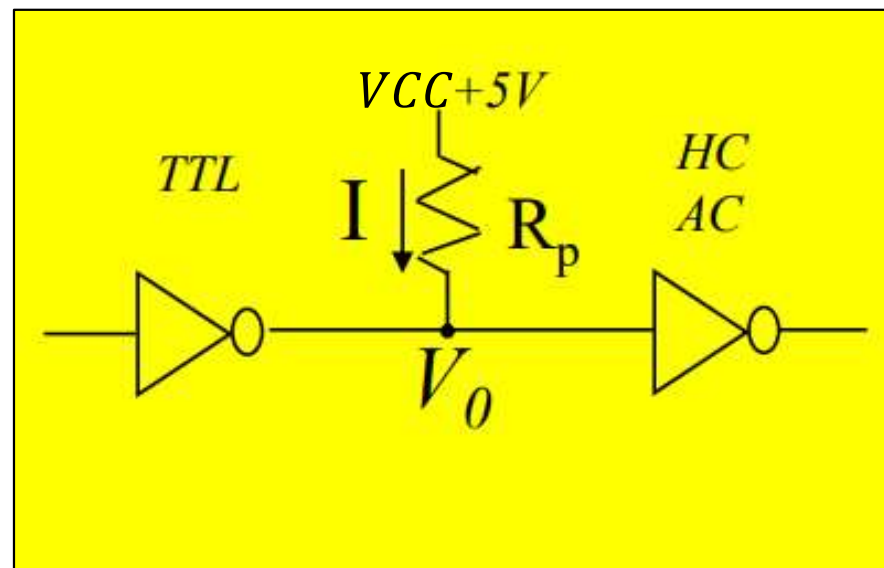
$$R_P > \frac{5V - 0.5V}{10mA} \Rightarrow R_P > 0.45k\Omega$$



Compatibility: 5V CMOS – TTL

The larger R_p is, the lower the power consumption in static conditions, but at the expense of the speed in dynamic condition.

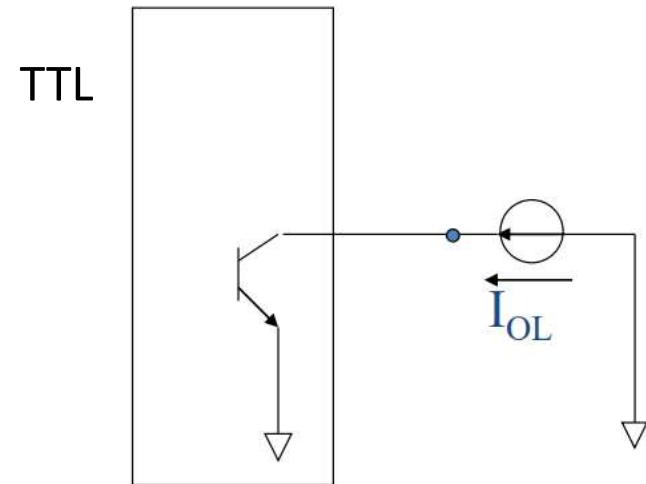
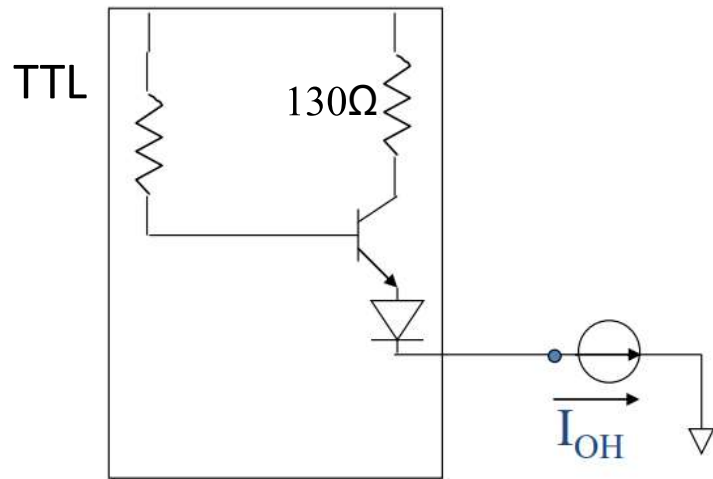
$$1.5k\Omega > R_p > 0.45k\Omega$$



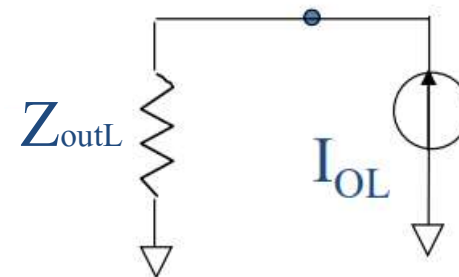
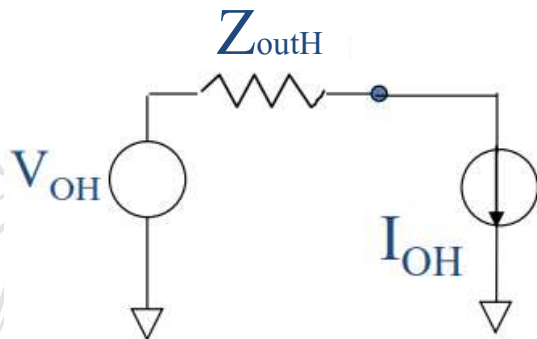
Output characteristics TTL

HIGH

LOW



Simplified equivalent circuits

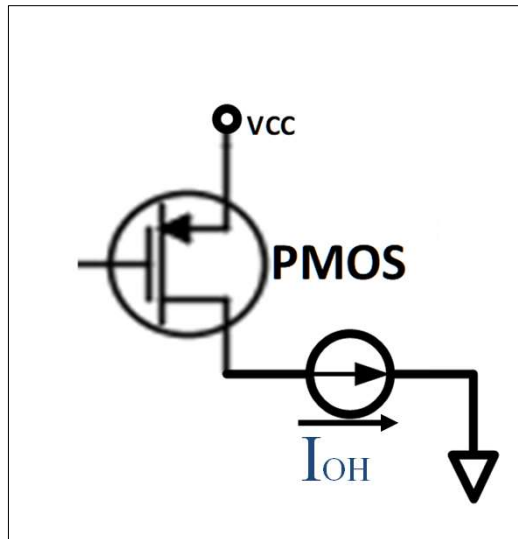


Output characteristics CMOS

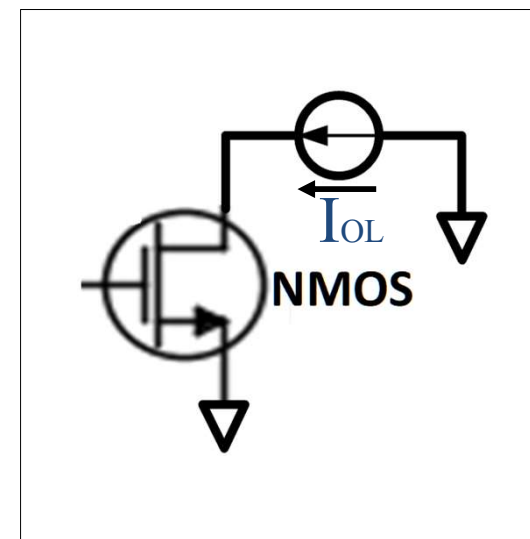
HIGH

LOW

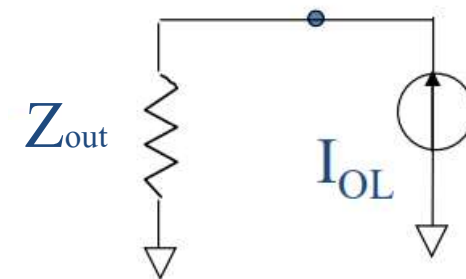
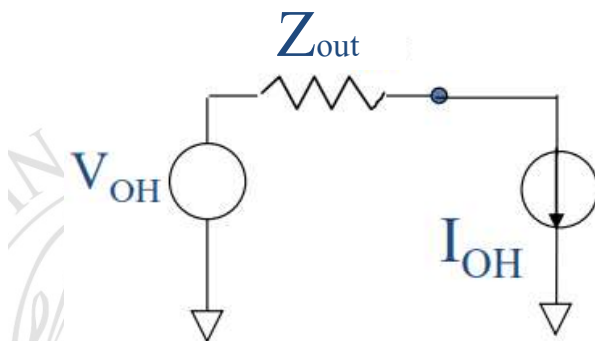
CMOS



CMOS



Simplified equivalent circuits





Max output currents

Max output current: the current corresponding to V_{OHmin} / V_{OLmax} guaranteed

	I_{OH} max (mA)	I_{OL} max (mA)
LS00	-0,4	+16
F00	-1	+20
F244	-15	+64
AHC00	-8	+8
LVC244	-24	+24

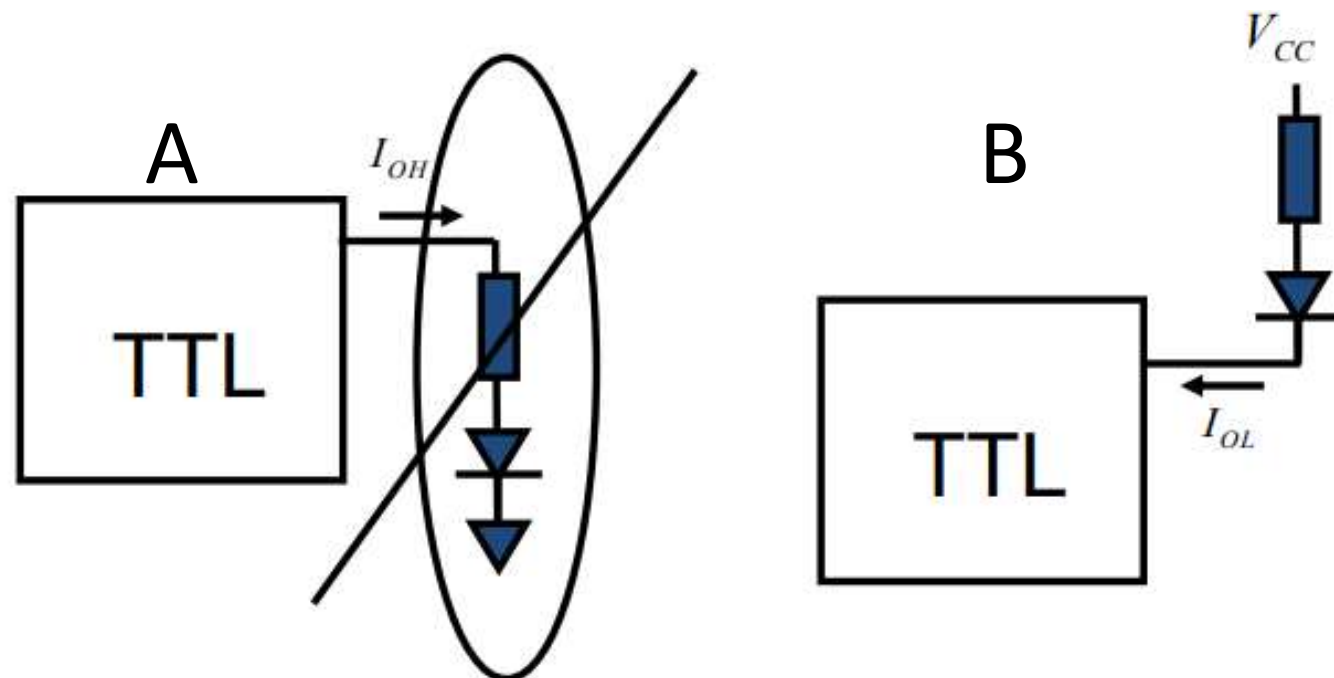
- Compared to simple gates (00), the buffers (244) guarantee higher output currents, especially in faster (F, LVC) logics
- CMOS devices drive similar currents at low and high levels
- Bipolar circuits drive higher currents at low level (strong asymmetry)



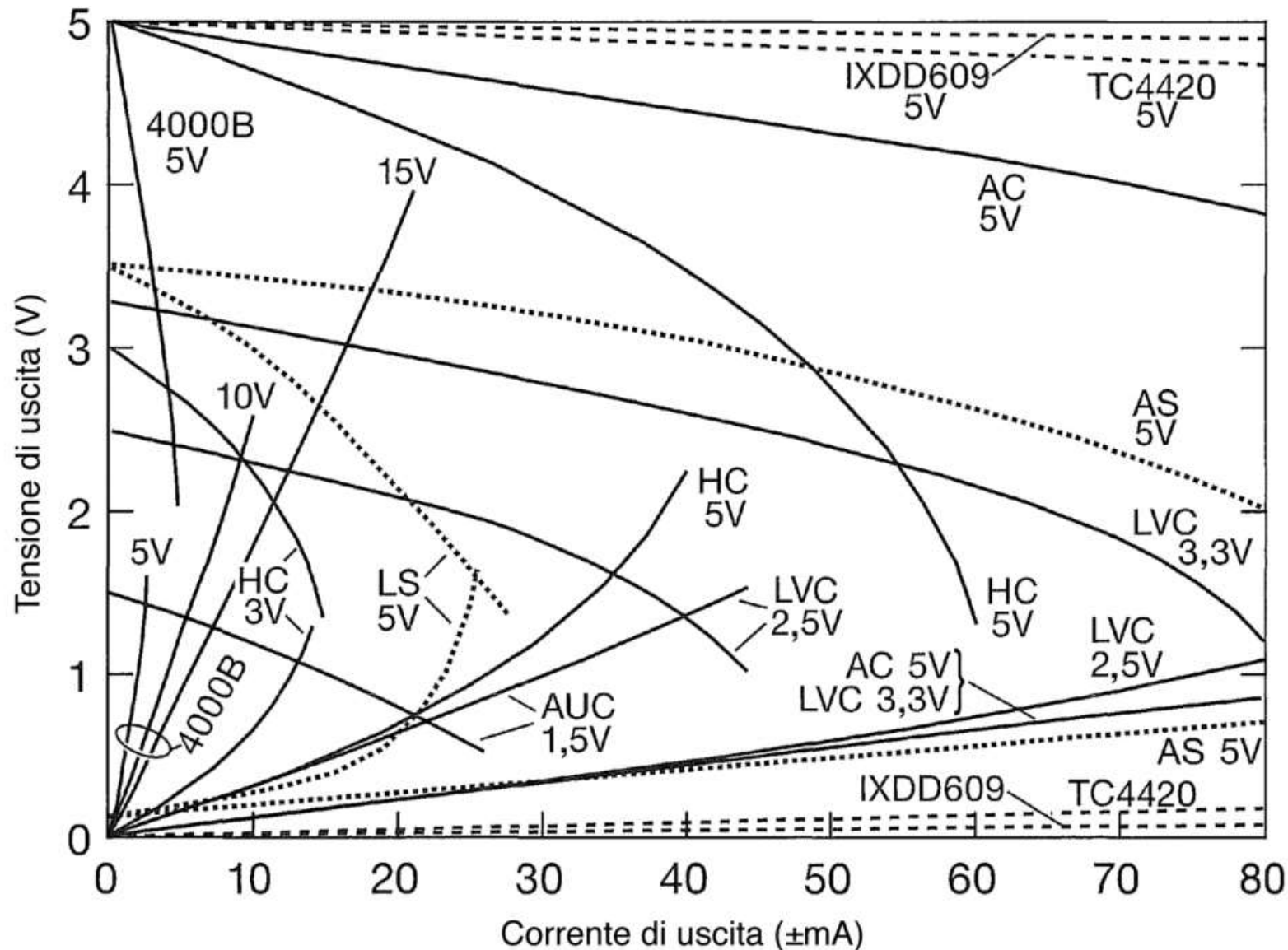
TTL output currents

The big difference between I_{OL} and I_{OH} is a peculiarity of the TTL, which pushes to exclude circuit solutions that require more current than the one guaranteed by the high level.

Ex: to switch-on a led, at least 15 mA are needed $\rightarrow$ the led is better switched-on by a low level than by a high level



General output characteristics



Horowitz-Hill: Arte dell'elettronica, Zanichelli (2018)

Logic Families

Family	Technology	Vcc Range	Key Characteristics
4000B	CMOS	3V – 18V	Ultra-low power, very wide voltage range, very slow.
LS	Bipolar (TTL)	5V only	Legacy standard. Good speed/power compromise.
AS	Bipolar (TTL)	5V only	Extremely fast switching, high power consumption.
HC	CMOS	2V – 6V	Mainstream general-purpose. High speed, low static power.
AC	CMOS	2V – 6V	Advanced speed, high output drive capability.
LVC	CMOS	1.65V – 3.6V	Modern 3.3V standard. 5V-tolerant inputs for level shifting.
AVC	CMOS	1.2V – 3.3V	Ultra-fast (< 2ns delay), optimized for high-speed routing.
IXDD609	BiCMOS	4.5V – 18V	6A output current. Extremely fast switching.
TC4420	BiCMOS	4.5V – 35V	9A output current. Optimized for high-power applications.



General output characteristics

- The V_{OH} , V_{OL} values specified in data sheets frequently refer to specific I_{OH} , I_{OL} values.
- When I_{out} is significant V_{OH} decreases and V_{OL} increases non-linearly.
- Of course:

The lower (resistance is) the load, the higher I_{out} and the higher the output voltage drop



Datasheets of digital ICs

The datasheets of logic ICs provide information about:

- Functionality
- Available Packages
- **Absolute maximum ratings**
- **Recommended operating conditions**
- **Electrical characteristics**
- **Switching characteristics**
- ...

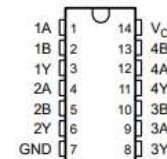
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SN74ALVC00
QUADRUPL 2-INPUT POSITIVE-NAND GATE
SCES115G-JULY 1997-REVISED AUGUST 2004

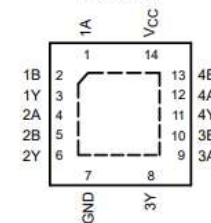
FEATURES

- Operates From 1.65 V to 3.6 V
- Max t_{pd} of 3 ns at 3.3 V
- ± 24 -mA Output Drive at 3.3 V
- Latch-Up Performance Exceeds 250 mA Per JESD 17
- ESD Protection Exceeds JESD 22
 - 2000-V Human-Body Model (A114-A)
 - 200-V Machine Model (A115-A)
 - 1000-V Charged-Device Model (C101)

D, DGV, NS, OR PW PACKAGE
(TOP VIEW)



RGY PACKAGE
(TOP VIEW)



DESCRIPTION/ORDERING INFORMATION

This quadruple 2-input positive-NAND gate is designed for 1.65-V to 3.6-V V_{CC} operation. The SN74ALVC00 performs the Boolean function $Y = \overline{A \cdot B}$ or $Y = \overline{A} + \overline{B}$ in positive logic.

ORDERING INFORMATION

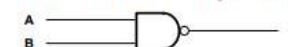
T_A	PACKAGE ⁽¹⁾		ORDERABLE PART NUMBER	TOP-SIDE MARKING
-40°C to 85°C	QFN - RGY	Tape and reel	SN74ALVC00RGYR	VA00
		Tube	SN74ALVC00D	
	SOIC - D	Tape and reel	SN74ALVC00DR	ALVC00
		Tape and reel	SN74ALVC00NSR	ALVC00
	TSSOP - PW	Tape and reel	SN74ALVC00PWR	VA00
	TVSOP - DGV	Tape and reel	SN74ALVC00DGVR	VA00

(1) Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

FUNCTION TABLE (each gate)

INPUTS		OUTPUT Y
A	B	
H	H	L
L	X	H
X	L	H

LOGIC DIAGRAM, EACH GATE (POSITIVE LOGIC)



Datasheets of digital ICs

SN74ALVC00 QUADRUPLE 2-INPUT POSITIVE-NAND GATE

SCES115G–JULY 1997–REVISED AUGUST 2004



ABSOLUTE MAXIMUM RATINGS⁽¹⁾

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V_{CC}	Supply voltage range	-0.5	4.6	V
V_I	Input voltage range ⁽²⁾	-0.5	4.6	V
V_O	Output voltage range ⁽²⁾⁽³⁾	-0.5	$V_{CC} + 0.5$	V
I_{IK}	Input clamp current	$V_I < 0$		-50 mA
I_{OK}	Output clamp current	$V_O < 0$		-50 mA
I_O	Continuous output current		±50	mA
	Continuous current through V_{CC} or GND		±100	mA
θ_{JA}	Package thermal impedance	D package ⁽⁴⁾		86
		DGV package ⁽⁴⁾		127
		NS package ⁽⁴⁾		76
		PW package ⁽⁴⁾		113
		RGY package ⁽⁵⁾		47
T_{stg}	Storage temperature range	-65	150	°C

- (1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.
- (3) This value is limited to 4.6 V maximum.
- (4) The package thermal impedance is calculated in accordance with JESD 51-7.
- (5) The package thermal impedance is calculated in accordance with JESD 51-5.

Absolute maximum ratings: if respected, the device will not be damaged



Datasheets of digital ICs

RECOMMENDED OPERATING CONDITIONS⁽¹⁾

		MIN	MAX	UNIT
V_{CC}	Supply voltage	1.65	3.6	V
V_{IH}	High-level input voltage	$V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$	$0.65 \times V_{CC}$	V
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.7	
		$V_{CC} = 2.7 \text{ V to } 3.6 \text{ V}$	2	
V_{IL}	Low-level input voltage	$V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$	$0.35 \times V_{CC}$	V
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	0.7	
		$V_{CC} = 2.7 \text{ V to } 3.6 \text{ V}$	0.8	
V_I	Input voltage	0	3.6	V
V_O	Output voltage	0	V_{CC}	V
I_{OH}	High-level output current	$V_{CC} = 1.65 \text{ V}$	-4	mA
		$V_{CC} = 2.3 \text{ V}$	-12	
		$V_{CC} = 2.7 \text{ V}$	-12	
		$V_{CC} = 3 \text{ V}$	-24	
I_{OL}	Low-level output current	$V_{CC} = 1.65 \text{ V}$	4	mA
		$V_{CC} = 2.3 \text{ V}$	12	
		$V_{CC} = 2.7 \text{ V}$	12	
		$V_{CC} = 3 \text{ V}$	24	
$\Delta t/\Delta v$	Input transition rise or fall rate		5	ns/V
T_A	Operating free-air temperature	-40	85	°C

(1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

Recommended operating conditions: if respected, the device will work as specified in the datasheet



Datasheets of digital ICs

ELECTRICAL CHARACTERISTICS

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS	V_{CC}	MIN TYP ⁽¹⁾ MAX	UNIT
V_{OH}	$I_{OH} = -100 \mu A$	1.65 V to 3.6 V	$V_{CC} - 0.2$	V
	$I_{OH} = -4 \text{ mA}$	1.65 V	1.2	
	$I_{OH} = -6 \text{ mA}$	2.3 V	2	
	$I_{OH} = -12 \text{ mA}$	2.3 V	1.7	
		2.7 V	2.2	
		3 V	2.4	
	$I_{OH} = -24 \text{ mA}$	3 V	2	
V_{OL}	$I_{OL} = 100 \mu A$	1.65 V to 3.6 V	0.2	V
	$I_{OL} = 4 \text{ mA}$	1.65 V	0.45	
	$I_{OL} = 6 \text{ mA}$	2.3 V	0.4	
	$I_{OL} = 12 \text{ mA}$	2.3 V	0.7	
		2.7 V	0.4	
	$I_{OL} = 24 \text{ mA}$	3 V	0.55	
I_I	$V_I = V_{CC}$ or GND	3.6 V	± 5	μA
I_{CC}	$V_I = V_{CC}$ or GND, $I_O = 0$	3.6 V	10	μA
ΔI_{CC}	One input at $V_{CC} - 0.6 \text{ V}$, Other inputs at V_{CC} or GND	3 V to 3.6 V	750	μA
C_i	$V_I = V_{CC}$ or GND	3.3 V	4.5	pF

(1) All typical values are at $V_{CC} = 3.3 \text{ V}$, $T_A = 25^\circ\text{C}$.

Electrical characteristics (static): Output voltages and Input currents vs I_O and V_{CC}



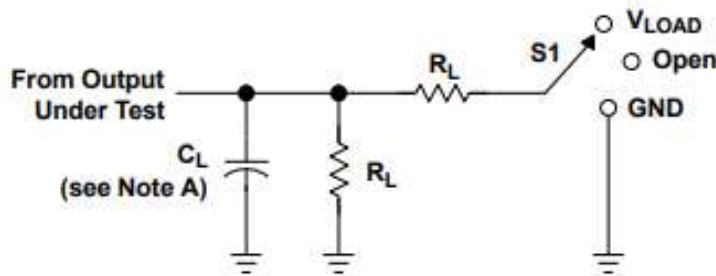
Datasheets of digital ICs

SWITCHING CHARACTERISTICS

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	$V_{CC} = 1.8\text{ V} \pm 0.15\text{ V}$		$V_{CC} = 2.5\text{ V} \pm 0.2\text{ V}$		$V_{CC} = 2.7\text{ V}$		$V_{CC} = 3.3\text{ V} \pm 0.3\text{ V}$		UNIT
			MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	
t_{pd}	A or B	Y	1	4.4	1	2.8	3.2		1	3	ns

PARAMETER MEASUREMENT INFORMATION



LOAD CIRCUIT

TEST	S1
t_{pd}	Open
t_{PLZ}/t_{PZL}	V_{LOAD}
t_{PHZ}/t_{PZH}	GND

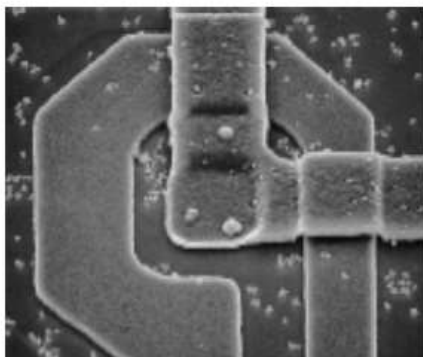
V_{CC}	INPUT		V_M	V_{LOAD}	C_L	R_L	V_A
	V_I	t_r/t_f					
$1.8\text{ V} \pm 0.15\text{ V}$	V_{CC}	$\leq 2\text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	1 k Ω	0.15 V
$2.5\text{ V} \pm 0.2\text{ V}$	V_{CC}	$\leq 2\text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	500 Ω	0.15 V
2.7 V	2.7 V	$\leq 2.5\text{ ns}$	1.5 V	6 V	50 pF	500 Ω	0.3 V
$3.3\text{ V} \pm 0.3\text{ V}$	2.7 V	$\leq 2.5\text{ ns}$	1.5 V	6 V	50 pF	500 Ω	0.3 V

Switching characteristics (dynamic): propagation delay vs V_{CC} (and in specified conditions)

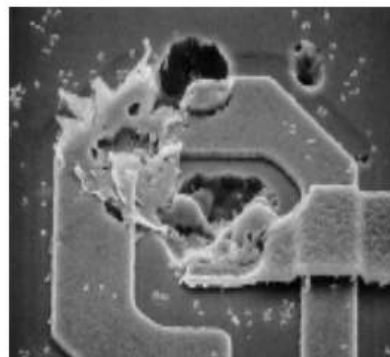
ElectroStatic Discharge (ESD)



- Due to transfer of electrostatic charge (up to 40 kV), originated by the contact between different materials
- The electrostatic charge is more easily accumulated in places with low ($< 20\%$) humidity
- ESD is a relevant problem, since the integrated circuits are built with smaller and smaller geometries, that might break (gate breakdown) at high voltage levels.



Pentium4 transistor



ESD effects

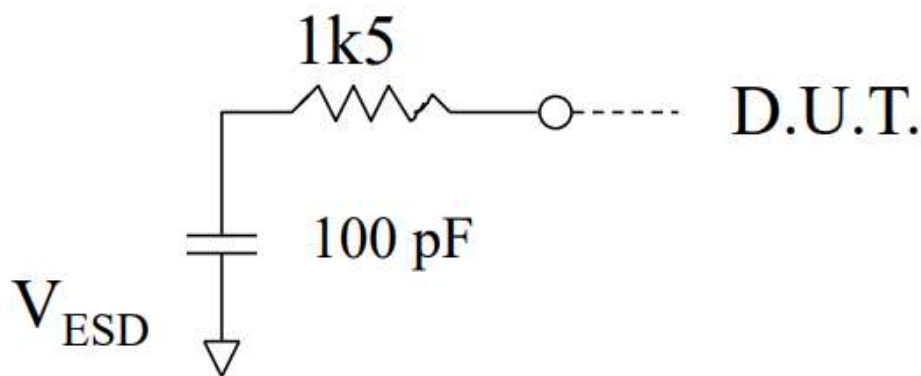


high power (X1200)

Figure 3.76. Scanning electron micrograph of a 6 amp MOSFET destroyed by 1kV charge on "human body equivalent" (1.5k in series with 100pF) applied to its gate. (Courtesy of Motorola, Inc.)

ESD: models, protective circuits, countermeasures

The charge transfer can be modeled by circuits like:

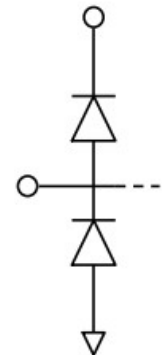


Human Body Model

This (simplified) model allows comparing the capability of different circuits to survive to the application of a given V_{ESD}

Protective circuits

- Polysilicon gates rather than metallic
- Transistors with high $V_{breakdown}$
- Input diodes + R_s
- Output diodes



Countermeasures

- Environment control
- Use antistatic materials
- Handling precautions

ESD protection rating for ICs

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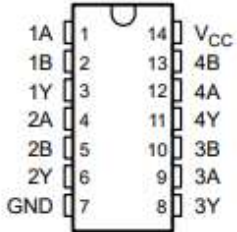
SN74ALVC00
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FEATURES

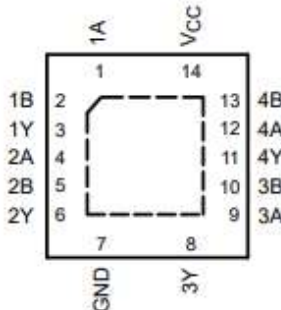
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**D, DGV, NS, OR PW PACKAGE
(TOP VIEW)**



**RGY PACKAGE
(TOP VIEW)**



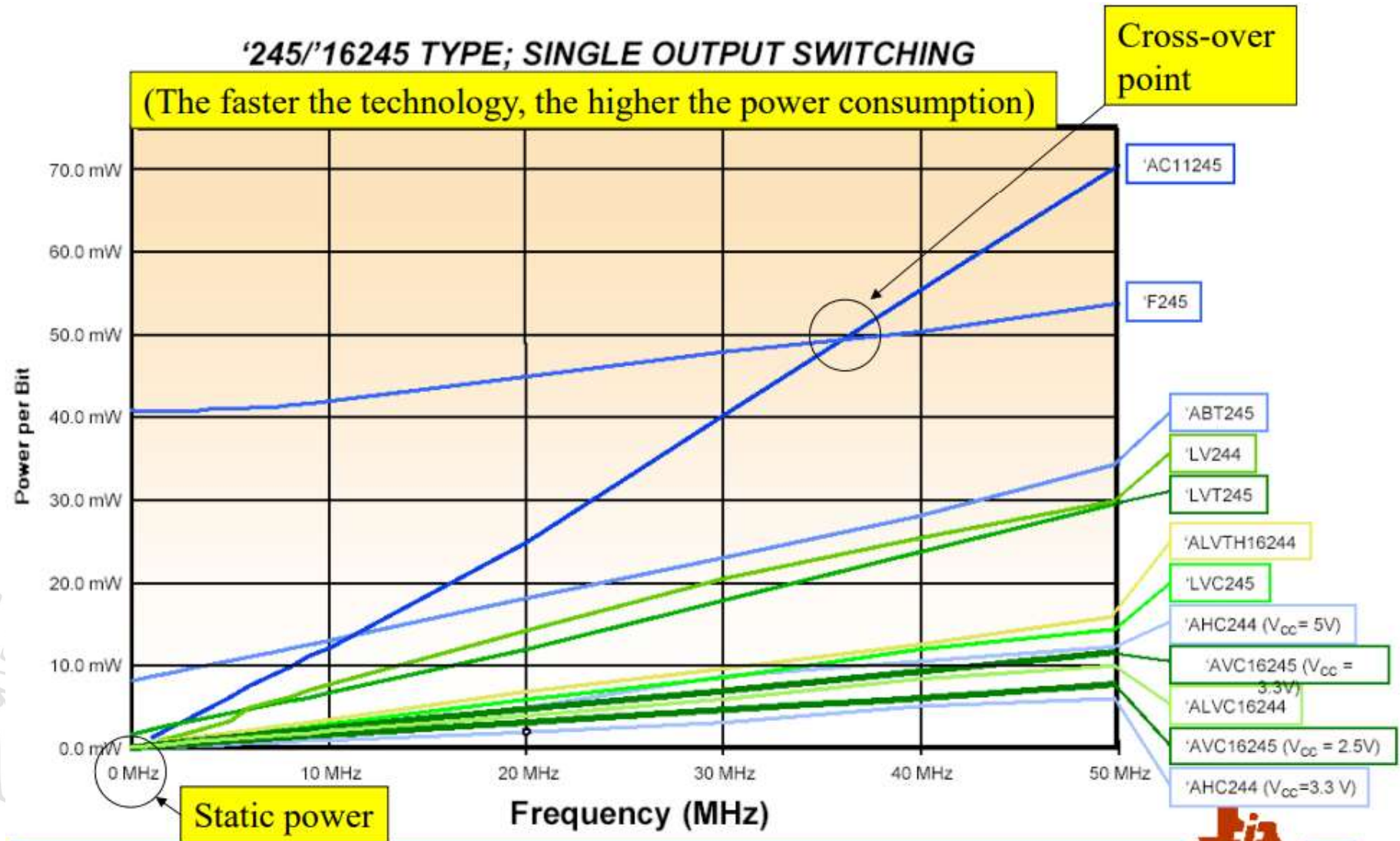


Digital integrated circuits

Dynamic power consumption



Logic families: Power vs Frequency





Dynamic power consumption in different logic families

- Significant power is consumed by bipolar families even at low frequencies (down to zero)
- CMOS family curves approximately start from zero
- BiCMOS family has an intermediate behavior
- Faster families have higher slopes
- At highest frequencies (beyond the cross-over point), fastest CMOS families may consume more than the fastest bipolar families



Dynamic power consumption in different logic families

In TTL circuits, the power consumption is due to:

- Bias currents of transistors (static consumption)
- Transient currents during the simultaneous conduction of totem-pole transistors (dynamic)

In CMOS circuits, the power consumption:

- is actually limited to switching (only dynamic)
- is mostly due to the charge/discharge of
 - parasitic capacitance in the integrated circuits
 - external load capacitance
- is noticeably lower in low voltage logic circuits (LV)

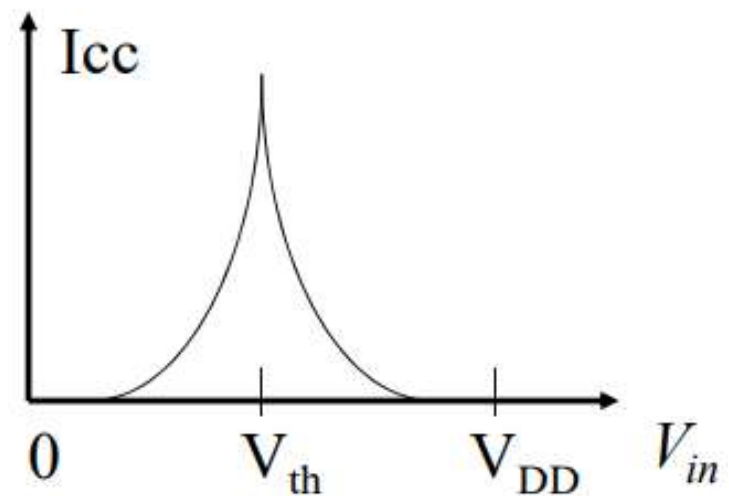
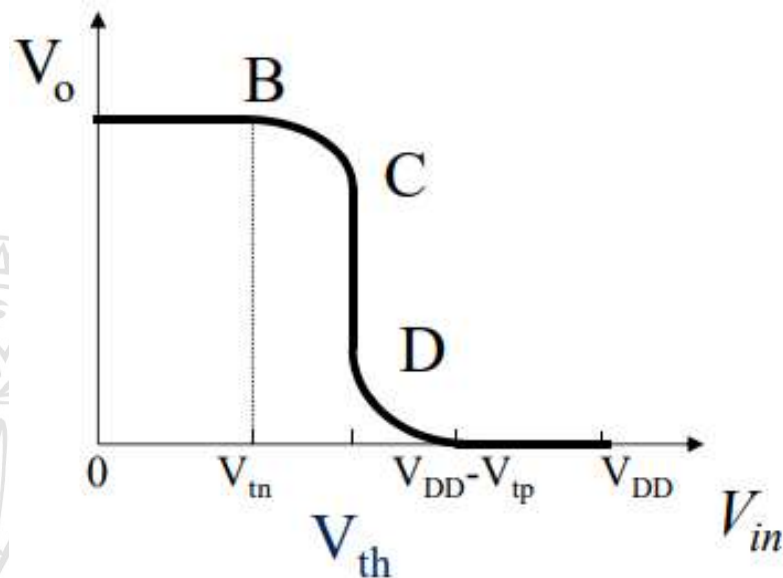


CMOS dynamic power

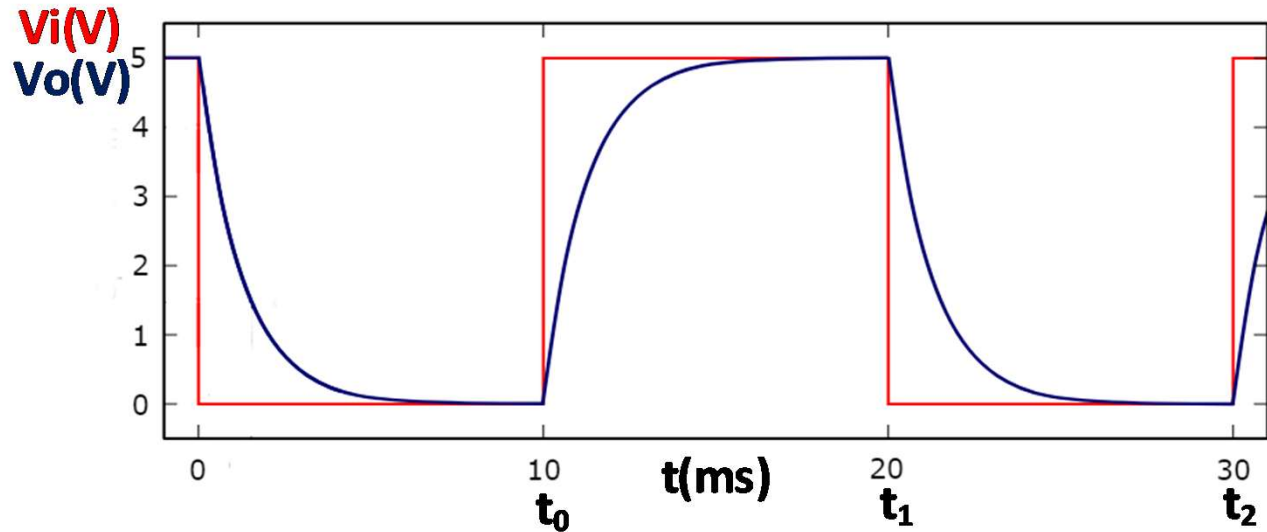
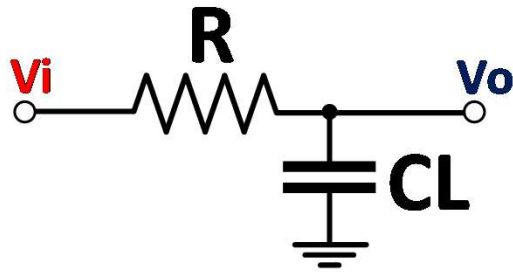
- The dynamic power consumption of CMOS is due to 2 mechanisms associated to the transistors:
 - Cross-conduction of n- and p-channel output transistors
 - Dissipation during charge/discharge of parasitic capacitances
- The second mechanism is largely more important (3-5 times higher)

CMOS power: cross-conduction of output transistors

By slowly varying the input voltage of an inverter, the measured output voltage, V_o , and the I_{cc} current, are like in the figure below. In particular, the max current is observed when $V_{in} = V_{th}$ (both transistors strongly conducting)



CMOS power: parasitic capacitance



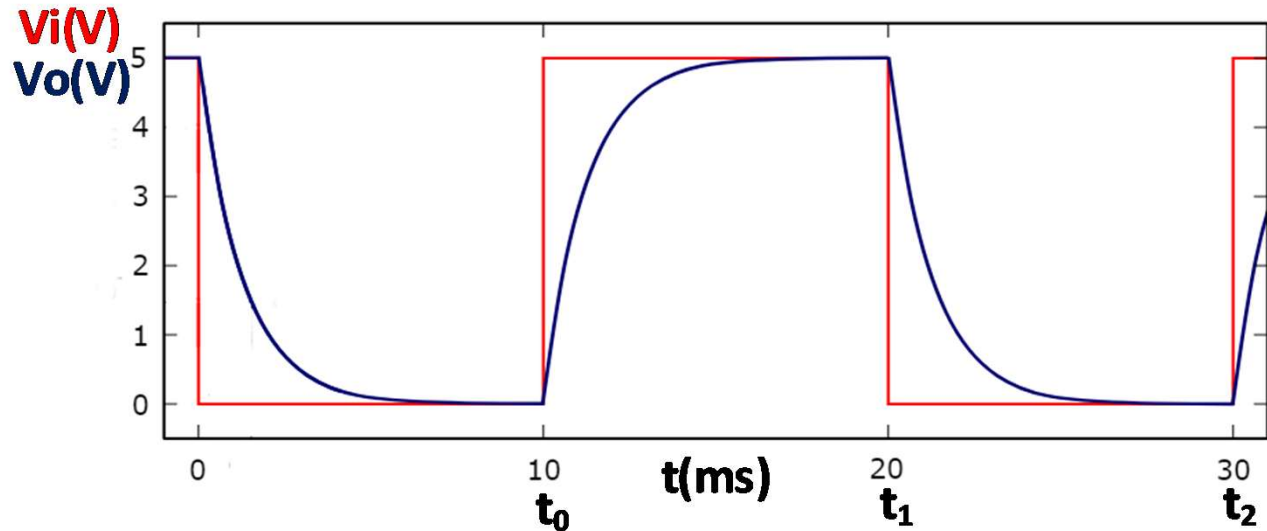
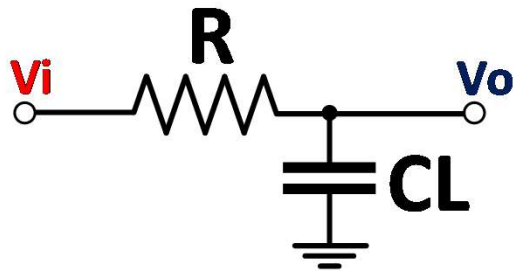
$$P = \frac{1}{T} \int_0^t i_0(t) \cdot v_0(t) dt$$

$$T = t_3 - t_0 \quad f_0 = \frac{1}{T}$$

$$i_0 = i_p = C_L \frac{dV}{dt} \quad \text{During capacitor charge}$$

$$i_0 = i_n = -C_L \frac{dV}{dt} \quad \text{During capacitor discharge}$$

CMOS power: parasitic capacitance



$$P = \frac{1}{T} \int_0^T i_0(t) \cdot v_0(t) dt = \frac{1}{T} \left[\int_{t_0}^{t_1} i_p(t) \cdot v_0(t) dt + \int_{t_1}^{t_2} i_n(t) \cdot v_0(t) dt \right]$$

$$P = \frac{1}{T} \left[\int_{t_0}^{t_1} C_L \frac{dv}{dt} \cdot v_0(t) dt + \int_{t_1}^{t_2} -C_L \frac{dv}{dt} \cdot v_0(t) dt \right] = \frac{C_L}{T} \left[\int_0^{V_{DD}} v_0 dv + \int_{V_{DD}}^0 -v_0 dv \right]$$

$$P = \frac{C_L}{T} \left[\frac{v_0^2}{2} \Big|_0^{V_{DD}} - \frac{v_0^2}{2} \Big|_{V_{DD}}^0 \right] = \frac{C_L}{T} \left[\frac{v_{DD}^2}{2} - \left(-\frac{v_{DD}^2}{2} \right) \right]$$

$$P = C_L \cdot v_{DD}^2 \cdot f_0$$

$$P_n = \sum_n (C_{Ln} \cdot f_{0n}) \cdot v_{DD}^2$$

CMOS power: parasitic capacitance

Power dissipated by a capacitor at clock frequency, f : $P = CV^2 f$

In CMOS ICs we should consider the:

- internal parasitic capacitance, C_{pd}
- load parasitic capacitance, C_L

NB: the power involved in this mechanism is 3-5 times higher than the one due to the simultaneous conduction of output transistors

C_{pd} (power dissipation capacitance) represents the capacitance of the «equivalent capacitor», i.e. the capacitor that, if substituted to the IC, would involve an equivalent power consumption





Typical C_{pd} values (from datasheets)

74AHC1G32 (single OR gate)

PARAMETER		TEST CONDITIONS	TYP	UNIT
C_{pd}	Power dissipation capacitance	No load, $f = 1 \text{ MHz}$	14	pF

74AC74 (dual FFD)

PARAMETER		TEST CONDITIONS	TYP	UNIT
C_{pd}	Power dissipation capacitance	$C_L = 50 \text{ pF}$, $f = 1 \text{ MHz}$	45	pF

74HC163 (4 bit counter)

PARAMETER		TEST CONDITIONS	TYP	UNIT
C_{pd}	Power dissipation capacitance	No load	60	pF

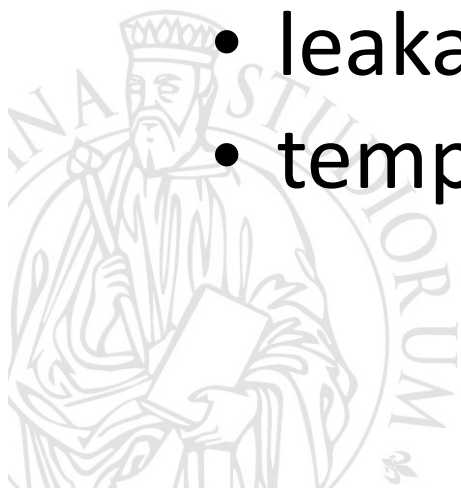
The C_{pd} value increases with the circuit complexity (but is lower in newer technologies: eg, 27 pF in 74LVC74)



Power consumption in complex systems

The dissipated power in complex digital systems (such as **FPGA**, **uC**, **DSP**, **uP**, **GPU**) mainly depends on:

- frequency
- supply voltage
- leakage currents
- temperature





Power management in complex systems

- Dynamic Voltage/Frequency scaling
- Clock gating
- Sub-system power partitioning
- Start&Stop
- Thermal throttling

